AgriPerceiver: A Parameter-Efficient Vision-Language Model for Structured Macroscopic Crop Phenotyping

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Abstract. Modern agriculture suffers from systemic crop yield instability, with plant pathogens contributing to 20–40% of global yield losses. Adapting generalist Vision-Language Models (VLMs) to produce structured, actionable phenotyping from crop images creates a scalability crisis at the intersection of computer vision and agricultural life sciences. We present **AgriPerceiver**, a lightweight VLM that frames this challenge as *structured report generation*: given a single leaf photograph, the model produces a schema-compliant JSON diagnostic report detailing disease identity, pathology type, severity score, symptom characterisation, and actionable steps. To process high-resolution visual imagery, the input is spatially decomposed to preserve fine-grained pathological features that are typically lost during standard resizing. Our central contribution is a *perception bridge* that mitigates visual token explosion by compressing visual tokens into just 128 learned latents (28.5× reduction) while critically preserving learned tile-position embeddings for spatial grounding. We employ a two-stage training curriculum: the bridge (∼391M parameters) is first aligned, followed by LoRA specialisation on Gemma-3-labelled structured annotations. Both the vision and language backbones remain strictly frozen throughout training to maintain parameter efficiency. Evaluated across nine metrics, AgriPerceiver achieves a composite score of **0.810** and 99.7% schema compliance on a held-out test set, and improves pathology-type macro-F1 by 1.9× over the strongest generalist baseline while training only 9.2% of its parameters, demonstrating the viability of parameter-efficient domain specialisation in life-sciences AI for structured knowledge extraction.

Keywords: Vision-Language Models · Agricultural Pathology · Perceiver Resampler · Life Sciences AI · LoRA · Structured Output Generation

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1 Introduction

Pathogen-induced crop losses account for 20–40% of global agricultural output annually [11], making plant disease diagnosis one of the highest-impact problems in life sciences and food security. Conventional workflows require trained agronomists to inspect field samples and write structured reports covering disease type, severity, observed symptoms, and treatment. The process is slow and resource-intensive, and cannot scale to the hundreds of millions of smallholder farms that most need it. Seen through the lens of life-sciences AI this is a multi-modal grounding problem: macroscopic visual phenotypes must be mapped to a structured biological ontology, pathogen taxonomy, severity index, and therapeutic action.

Vision-language models (VLMs) offer a principled path toward automating this diagnosis, yet adapting a general-purpose VLM to agricultural pathology exposes three limitations. *First*, standard visual token budgets are designed for natural images, not the high-resolution, fine-grained symptom patterns (*e.g.* fungal lesions, bacterial water-soaking, nutrient-deficiency chlorosis) that separate disease classes. Naively raising the token budget to preserve these details incurs a quadratic attention cost that quickly exceeds commodity hardware. *Second*, autoregressive VLM output is free-form text, incompatible with the structured JSON schemas required by agricultural decision-support APIs. *Third*, state-of-the-art VLMs carry billions of parameters trained on web-scale corpora and cannot be specialised affordably on low-compute infrastructure.

We address these limitations with **AgriPerceiver**, contributing:

1. A *perception bridge*: AnyRes spatial tiling, learned tile-position embeddings, an MLP projector, and a two-block **Perceiver Resampler** that compresses 3,645 visual tokens into 128 latents (28.5×), reducing per-layer attention cost by 27.5× (Sect. 4.6) while retaining the spatial fidelity needed for fine-grained symptom analysis.
2. A *two-stage training curriculum*: **bridge-only alignment pretraining** followed by LoRA-based specialisation, keeping both frozen backbones (∼4.22B parameters) out of the gradient path, so only ∼426M parameters (∼9.2% of 4.6B) are updated in Stage 2.
3. An *automated data pipeline and rigorous evaluation framework*: we leverage Gemma-3-12B-IT [5] to label 117K agricultural images into a schema-compliant, seven-field JSON format. Model assessment is then conducted across a comprehensive nine-metric suite covering five distinct quality dimensions, together with a per-class, calibration, and complexity analysis.

Across all five evaluation axes, AgriPerceiver attains a composite score of 0.810 versus 0.615 for the strongest generalist baseline, a 1.9× improvement in pathology-type macro-F1 over the strongest baseline (up to 4.8× over the weakest), and a 3.3× reduction in severity error, while remaining trainable on a single GPU.

2 Related Work

2.1 Vision-Language Models and Structured Output

Generalist VLMs such as LLaVA [9], InternVL [3], and Qwen2-VL [12] couple a frozen vision encoder to a large language model through a lightweight connector and are strong at visual question answering and captioning. Their objective, however, optimises free-form conversational text; when prompted for machine-readable output they produce syntactically valid but semantically shallow structures, as our experiments confirm (Sect. 6.2). Reliable structured scientific output, where a downstream API consumes a fixed schema, remains an open challenge that prompt engineering alone does not solve. AgriPerceiver instead treats the schema as a first-class training target, so that structural compliance and field-level accuracy are learned jointly.

2.2 Visual Token Compression

Visual token count grows with input resolution and self-attention over those tokens scales quadratically, creating a bottleneck for high-resolution scientific imagery. The Perceiver [7] and Flamingo's Perceiver Resampler [2] address this with a small set of learned latent queries that cross-attend to the full visual sequence, decoupling downstream cost from input length. Recent VLMs (*e.g.* InternVL) instead employ dynamic high-resolution tiling for general scenes. AgriPerceiver combines both ideas and adapts them to pathology: it tiles at high resolution to expose sub-millimetre lesion detail, then compresses aggressively through a Perceiver Resampler, while injecting learned tile-position embeddings so that spatial provenance survives compression, a property we show aids free-text localisation of symptoms (Sect. 6.5).

2.3 Parameter-Efficient Adaptation

Full fine-tuning of multi-billion-parameter backbones is expensive and risks catastrophic forgetting. Parameter-efficient fine-tuning, most prominently LoRA [6], injects low-rank update matrices into frozen weights, updating a small fraction of parameters; quantised variants such as QLoRA [4] further cut the memory footprint of both training and the teacher models used for annotation. AgriPerceiver pushes this paradigm to its limit for structured generation: both the SigLIP [13] vision encoder and the Phi-3-mini [1] language model stay strictly frozen, and only the perception bridge and rank-32 LoRA adapters, together 9.2% of the system, receive gradients.

2.4 AI for Agricultural Pathology

PlantVillage [10] established the foundation for deep-learning disease classification, but its legacy rests on scalar, single-label predictions over curated, controlled-background imagery. Domain-specialised multimodal pretraining has proven

effective at molecular and clinical scales (for example BiomedCLIP [14]), yet macroscopic agricultural pathology remains comparatively underserved. Moving from single-label classification to multi-dimensional diagnostic reporting (identity, type, severity, symptoms, reasoning, and actions) requires high-resolution imagery under tight compute constraints and a structured report rather than a class index. AgriPerceiver targets exactly this gap.

3 Problem Formulation

We cast macroscopic crop phenotyping as *conditional structured sequence generation*. Given a single leaf image $I \in \mathbb{R}^{H\times W\times 3}$ of arbitrary resolution and a fixed instruction prompt $\mathbf{x}_{\text{txt}}$, the model must produce a token sequence $\mathbf{y} = (y_1, \ldots, y_L)$ that deserialises into a JSON object R conforming to a fixed seven-field schema $\mathcal{S}$:

$$R = \{\,\text{diagnosis, type, severity, confidence, symptoms, reasoning, actions}\,\}. \tag{1}$$

Here type $\in \mathcal{T}$ is a categorical pathology label over $|\mathcal{T}|$=6 classes, *severity* and *confidence* are continuous values in $[0, 1]$, and the remaining fields are free-form natural language. Let f_ϕ denote the frozen vision encoder, g_θ the trainable perception bridge, and p_ψ the language model with LoRA parameters ψ. The model factorises the report autoregressively,

$$p(\mathbf{y} \mid I, \mathbf{x}_{\text{txt}}) = \prod_{t=1}^{L} p_\psi\big(y_t \mid y_{<t},\, g_\theta(f_\phi(I)),\, \mathbf{x}_{\text{txt}}\big), \tag{2}$$

and is trained to maximise the likelihood of teacher-provided reference reports. A prediction is deemed *schema-compliant* if $\mathbf{y}$ parses as valid JSON and instantiates all seven fields of $\mathcal{S}$ with type-correct values. This formulation unifies classification (the type field), regression (severity), and language generation (symptoms, reasoning, actions) under a single decoding objective, and motivates the multi-axis evaluation of Sect. 6.1.

4 Method

4.1 Architecture Overview

AgriPerceiver (Fig. 1) processes an input leaf image through four sequential stages: (1) a frozen SigLIP-SO400M [13] vision encoder, (2) a trainable *perception bridge* (TileEmbeddings → VisionProjector → PerceiverResampler), (3) a splice-and-forward multimodal fusion mechanism, and (4) a Phi-3-mini-128k [1] language model with LoRA adapters in Stage 2. Phi-3-mini's 128K context window accommodates long structured JSON outputs without truncation; its compact 3.8B parameter footprint keeps the full inference stack within a single-GPU 24 GB VRAM budget.

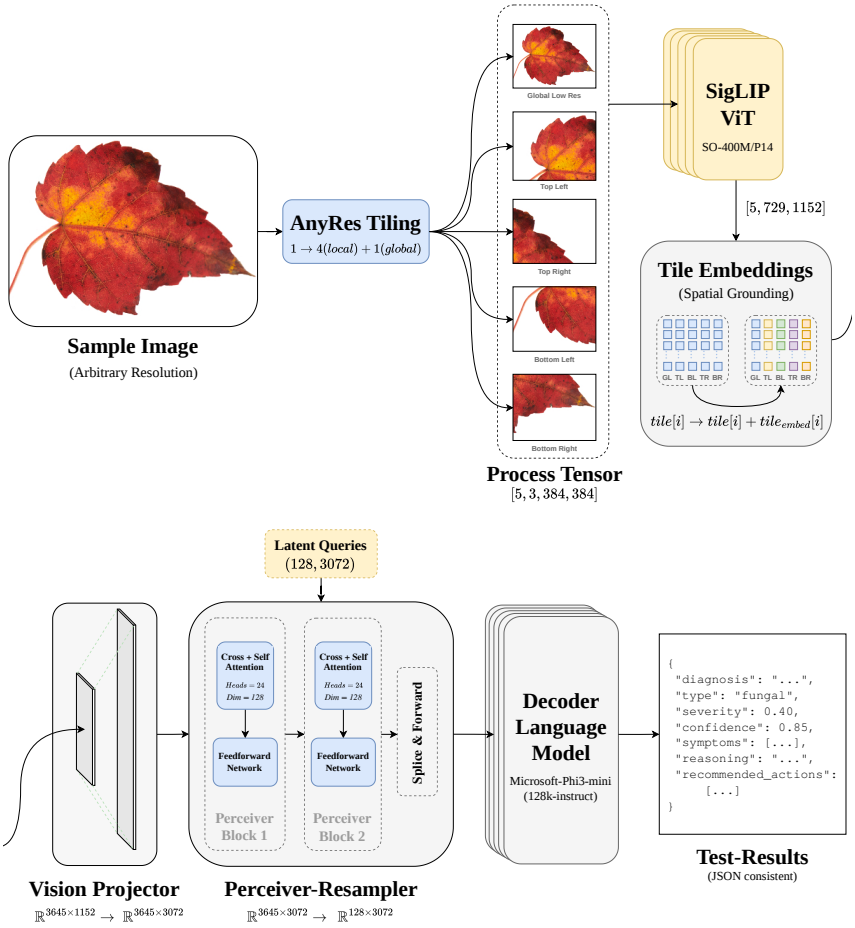


Fig. 1. AgriPerceiver architecture, split into two rows that read left-to-right, the lower row continuing from the upper. **(a) Tiling and encoding:** the input is decomposed into 5 AnyRes tiles (global view plus four quadrants, ≈2× effective resolution); a frozen SigLIP encodes them and learned tile embeddings restore spatial provenance. **(b) Compression and generation:** a two-block Perceiver Resampler (each block applies cross-attention to the visual features, latent self-attention, and a GLU-FFN) compresses 3,645 tokens to 128 latents (28.5×), which are spliced into a frozen Phi-3-mini to drive structured JSON generation.

4.2 AnyRes Spatial Tiling

Each input image $I \in \mathbb{R}^{H\times W\times 3}$, regardless of native resolution, is decomposed into five tiles: four equal quadrant crops (top-left, top-right, bottom-left, bottom-right) and one globally resized view, each rescaled to 384×384. The tile tensor $\mathbf{T} \in \mathbb{R}^{5\times3\times384\times384}$ is processed in parallel by SigLIP, yielding patch features $\mathbf{V} \in \mathbb{R}^{5\times729\times1152}$ ($729 = 27\times27$ patches per tile, $d_v = 1152$). Quadrant crops

provide approximately 2× effective resolution for fine-grained lesion analysis; the global tile preserves holistic leaf context.

4.3 Perception Bridge

Tile Embeddings. A learned parameter matrix $\mathbf{E} \in \mathbb{R}^{5\times 1152}$ assigns each tile a distinct spatial identity. For tile t, embedding $\mathbf{e}_t$ is broadcast and added to all 729 patch tokens:

$$\widetilde{\mathbf{V}}_{t,p} = \mathbf{V}_{t,p} + \mathbf{e}_t, \quad t \in \{1,\dots,5\},\ p \in \{1,\dots,729\}. \tag{3}$$

This encoding enables the downstream modules to recover the spatial origin of each patch without modifying the frozen encoder.

Vision Projector. The enriched features are reshaped to $\widetilde{\mathbf{V}} \in \mathbb{R}^{3645\times 1152}$ ($3645 = 5 \times 729$) and projected into the LLM dimension $d_l = 3072$ via a two-layer MLP with GELU activation:

$$\mathbf{V}' = \mathrm{MLP}_\theta(\widetilde{\mathbf{V}}) \in \mathbb{R}^{3645\times 3072}. \tag{4}$$

Perceiver Resampler. A set of $N = 128$ learned latent queries $\mathbf{Q} \in \mathbb{R}^{128\times 3072}$ iteratively attend to $\mathbf{V}'$ through $D = 2$ PerceiverBlocks. Each block applies scaled dot-product attention

$$\mathrm{Attn}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \mathrm{softmax}\Big(\tfrac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_h}}\Big)\mathbf{V}, \qquad d_h = d_l/H, \tag{5}$$

in a cross-attention, self-attention, feed-forward sequence with RMSNorm and residual connections:

$$\begin{aligned}
\mathbf{Z}^{(d)} &= \mathbf{Z}^{(d-1)} + \mathrm{CrossAttn}\Big(\mathrm{RMSNorm}(\mathbf{Z}^{(d-1)}),\ \mathbf{V}'\Big),\\
\mathbf{Z}^{(d)} &= \mathbf{Z}^{(d)} + \mathrm{SelfAttn}\Big(\mathrm{RMSNorm}(\mathbf{Z}^{(d)})\Big),\\
\mathbf{Z}^{(d)} &= \mathbf{Z}^{(d)} + \text{GLU-FFN}\Big(\mathrm{RMSNorm}(\mathbf{Z}^{(d)})\Big),
\end{aligned} \tag{6}$$

with $H = 24$ attention heads, where in the cross-attention the queries derive from the latents $\mathbf{Z}$ and the keys/values from the visual features $\mathbf{V}'$. The GLU-FFN is a gated linear unit, $\text{GLU-FFN}(\mathbf{u}) = \big(\mathrm{GELU}(\mathbf{u}\mathbf{W}_g) \odot \mathbf{u}\mathbf{W}_v\big)\mathbf{W}_o$ with a 4× hidden expansion. Initialising $\mathbf{Z}^{(0)} = \mathbf{Q}$, the final output $\mathbf{Z} \in \mathbb{R}^{128\times 3072}$ achieves 3,645 → 128 token compression (28.5×) with no reduction in the capacity available to the downstream LLM.

4.4 Splice-and-Forward Fusion

The 128 compressed latents replace the `<image>` placeholder in the text prompt:

$$\mathbf{X} = \big[\mathbf{x}_{\mathrm{pre}};\ \underbrace{\mathbf{Z}}_{128\,\text{tokens}};\ \mathbf{x}_{\mathrm{post}}\big], \tag{7}$$

where $\mathbf{Z}$ acts as the visual prompt representation. Phi-3's causal transformer then processes this hybrid sequence with `use_cache=False`.

4.5 Two-Stage Training Curriculum

Stage 1: Alignment. Only the perception bridge (∼391M parameters: TileEmbeddings + VisionProjector + PerceiverResampler; see Appendix A) is trained on generic image–caption pairs from 117K agricultural images using standard causal LM cross-entropy loss. Both SigLIP and Phi-3 are frozen throughout. Training runs for 2 epochs (14.7K gradient steps) with batch size 16, learning rate 10^{-4} (AdamW), and bfloat16 precision.

Stage 2: Specialisation. LoRA adapters (r=32, α=64, dropout 0.1) are attached to Phi-3's `qkv_proj`, `o_proj`, `gate_up_proj`, `down_proj`, and `up_proj` modules (∼35M parameters). The bridge and LoRA (∼426M total, ≈9.2% of the 4.6B system) are jointly fine-tuned on 96,239 structured JSON samples for 3 epochs (37.9K steps, batch size 8, gradient accumulation k=2) using a per-sample weighted cross-entropy objective:

$$\mathcal{L} = \frac{1}{|B|} \sum_{i \in B} w_i \cdot \mathrm{CE}(\hat{\mathbf{y}}_i, \mathbf{y}_i), \tag{8}$$

where $w_i \in [0, 1]$ is derived from Gemma-3's per-sample generation confidence, providing a curriculum effect that up-weights high-confidence structured labels.

4.6 Attention Complexity Analysis

The primary motivation for the Perceiver Resampler is to decouple attention cost from input resolution, which is essential when small lesions, pustules and early water-soaking must remain resolvable. Let T_v be the number of visual tokens, N the number of latents, and d the model width. Standard self-attention over the full sequence forms a $T_v \times T_v$ attention map and costs

$$\mathcal{F}_{\text{full}} = 4\, T_v^2\, d \quad \text{(FLOPs per layer, projections excluded)}, \tag{9}$$

scaling *quadratically* in T_v. The Perceiver instead lets N latents cross-attend to the T_v tokens and then self-attend among themselves,

$$\mathcal{F}_{\text{perc}} = \underbrace{4\, N\, T_v\, d}_{\text{cross-attn}} + \underbrace{4\, N^2\, d}_{\text{latent self-attn}}, \tag{10}$$

which is *linear* in T_v. For our configuration (T_v=3,645, N=128, d=3072), this yields $\mathcal{F}_{\text{full}}$=163.3 GFLOP versus $\mathcal{F}_{\text{perc}}$=5.93 GFLOP per layer, a 27.5× reduction (Table 1), and shrinks the per-head attention map from 13.29M to 0.47M entries. The gap widens with every additional tile: doubling the tile count to ten doubles $\mathcal{F}_{\text{perc}}$ but quadruples $\mathcal{F}_{\text{full}}$, to 11.7 and 653.0 GFLOP respectively, a 56.0× reduction. Because the latent count is fixed, the 128 latents furnish a constant-size "visual summary" whose downstream cost in the language model is invariant to how many AnyRes tiles we add, ensuring predictable inference latency as resolution scales.

Table 1. Attention cost per layer, full self-attention over the visual sequence versus the Perceiver Resampler, at the deployed configuration (T_v=3,645 tokens, N=128 latents, d=3072). FLOPs from Eqs. (9) and (10); projection terms excluded for both.

	Full self-attn.	**Perceiver**	**Reduction**
Attention scaling	$O(T_v^2 d)$	$O(NT_v d)$	—
Attn. map / head (entries)	13.29 M	0.47 M	28.5×
Attn. FLOPs / layer	163.3 G	5.93 G	27.5×
Downstream visual tokens	3,645	128	28.5×

5 Automated Data Pipeline

We curate 117,635 agricultural leaf images and generate structured labels using Gemma-3-12B-IT [5] operating under 4-bit NF4 quantisation [4] with an eight-shot structured-JSON exemplar prompt. Each label encodes the seven-field JSON schema of Eq. (1): *diagnosis*, *type* (fungal/bacterial/viral/pest/deficiency/unknown), *severity* $\in [0, 1]$, *confidence* $\in [0, 1]$, *symptoms*, *reasoning*, and *recommended_actions* (see Appendix B for a full example).

Crucially, to ensure dataset quality, generation confidence is derived directly from Gemma-3's output-token transition probabilities (the geometric mean of exp(transition_score)). This provides a calibrated signal grounded in the model's inherent generative uncertainty, entirely bypassing the unreliability of its self-reported `confidence` field.

Applying a strict filtering threshold based on this metric (τ=0.5), 101,301 samples (86%) pass our quality checks, yielding 96,239 training instances and a 5,062-sample held-out test set. The training samples are stratified by the same confidence signal into three curriculum buckets that set the per-sample training weight w_i of Eq. (8): hard, $[0.4, 0.6)$, w_i=0.5 (∼9% of samples); medium, $[0.6, 0.8)$, w_i=0.8 (∼35%); and easy, $[0.8, 1.0]$, w_i=1.0 (∼56%). The test set mirrors the training class balance (support counts in Table 3), with fungal and unknown together accounting for 60.4% of samples.

6 Experiments

6.1 Evaluation Framework

We measure model quality along nine metrics across five axes. **Structural**: JSON validity (%) and schema compliance (%). **Classification**: pathology-type macro-F1 and diagnosis fuzzy-match. Macro-F1 averages the per-class F1 uniformly,

$$F_1^{\text{type}} = \frac{1}{|\mathcal{T}|} \sum_{c \in \mathcal{T}} \frac{2\, P_c R_c}{P_c + R_c}, \tag{11}$$

so that rare classes count equally. Diagnosis fuzzy-match uses a token-Jaccard criterion: for predicted and reference name token sets A, B, the sample counts

Table 2. Main results on the 5,062-sample held-out test set. ↓: lower is better. AgriPerceiver (4.6B) trains only bridge+LoRA (∼426M, 9.2%); both backbones frozen; our model is set in bold. Columns: JSON validity and schema compliance (%); pathology-type macro-F1 and diagnosis fuzzy-match (FM); severity MAE and Pearson r; symptom BERTScore F1; and the composite of Eq. (12).

Model	**Structural**		**Classification**		**Regression**		**Sem.**	**Comp.**
	JSON	Schema	F1	FM	MAE↓	r	BERT	
LLaVA-NeXT-7B [8]	99.9	99.9	0.134	0.001	0.344	0.226	0.847	0.590
InternVL2-8B [3]	100.0	100.0	0.336	0.005	0.385	0.624	0.848	0.611
Qwen2-VL-7B [12]	99.9	99.9	0.287	0.001	0.224	0.666	0.830	0.615
AgriPerceiver	99.7	99.7	**0.645**	**0.486**	**0.067**	**0.855**	**0.921**	**0.810**

as correct when $J(A,B) = |A \cap B|/|A \cup B| \geq 0.6$. **Regression**: severity mean absolute error (MAE) and Pearson r. **Semantic**: BERTScore F1 [15] for the symptoms, reasoning, and recommended-actions fields. **Calibration**: Expected Calibration Error over M=10 equal-width confidence bins (lower is better), $\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \left|\overline{\mathrm{acc}}(B_m) - \overline{\mathrm{conf}}(B_m)\right|$, with N=5,062. These are aggregated into a single composite score:

$$\begin{aligned}\mathcal{C} = {} & 0.20\, F_1^{\mathrm{type}} + 0.15\, F_{\mathrm{diag}} + 0.15\, B_{\mathrm{sym}} + 0.10\,(1 - \mathrm{MAE}) \\ & + 0.10\, B_{\mathrm{reas}} + 0.10\, B_{\mathrm{act}} + 0.10\, J_{\mathrm{val}} + 0.05\,(1 - \mathrm{ECE}) \\ & + 0.05\, S_{\mathrm{cmp}}, \end{aligned} \tag{12}$$

where $B_{\cdot}$ are BERTScore F1 values, J_{val} is JSON validity, and S_{cmp} is schema compliance; all components lie in $[0,1]$ (higher is better). The weighting places the highest single weight on pathology type, the primary agronomic decision signal, while the three semantic BERTScore terms collectively receive the largest share (0.35), reflecting the report's role as a human-facing document. The ranking is robust to substantial reweighting: doubling and renormalising the structural, classification, or semantic weights in turn yields 0.835, 0.749, and 0.838 for AgriPerceiver against a best-baseline 0.663, 0.503, and 0.677, a minimum margin of +0.161.

6.2 Main Results

Table 2 compares AgriPerceiver against three general-purpose VLMs of comparable or larger scale on the 5,062-sample held-out test set. All baselines receive the identical structured-JSON system prompt and are evaluated with the same metric pipeline. AgriPerceiver achieves near-perfect structural compliance (99.7%). Severity regression yields Pearson $r = 0.855$ and MAE = 0.067 on the $[0,1]$ scale (RMSE = 0.125). Both lie within useful bounds for severity triage, and the MAE is 3.3× lower than that of the best baseline (Qwen2-VL-7B; MAE = 0.224). Pathology-type macro-F1 of 0.645 is 1.9× the best generalist baseline (InternVL2,

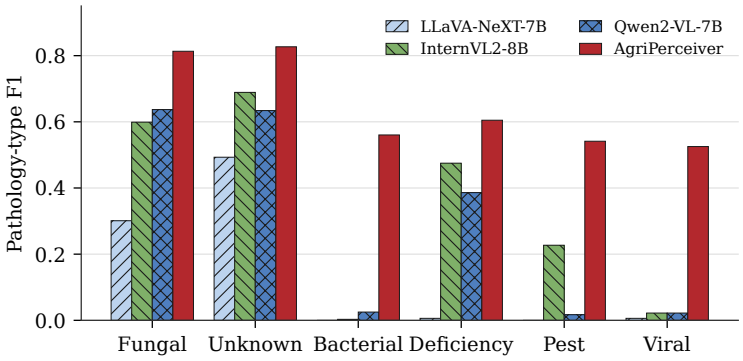


Fig. 2. Per-class pathology-type F1 on the held-out test set. Generalist baselines succeed only on the large, visually distinctive fungal and unknown classes and collapse on bacterial, pest, and viral pathologies; AgriPerceiver (rightmost, unhatched bar in each group) is the only model usable across all six classes. Bars are distinguished by hatch pattern as well as by shade, so the figure remains readable in greyscale. Baseline values follow the identical structured-JSON prompt.

0.336) and 4.8× the weakest (LLaVA-NeXT, 0.134). The overall composite of 0.810 exceeds the strongest baseline (Qwen2-VL, 0.615) by +0.195 (+32%). Baselines exhibit near-total failure on bacterial (F1 $\leq$ 0.025), pest (F1 $\leq$ 0.017), and viral (F1 $\leq$ 0.022) classes, and LLaVA-NeXT collapses to predicting "unknown" for 98.7% of samples, a degenerate strategy examined further in Sect. 6.3.

6.3 Per-Class Analysis

Fig. 2 decomposes pathology-type F1 by class. The generalist baselines are competitive only on *fungal* and *unknown* (the two largest and most visually distinctive classes) and collapse to near-zero F1 on bacterial, pest, and viral pathologies, whose discriminative cues (leaf-margin water-soaking, pest frass, viral mosaic) are subtle and not salient under generic visual prompting. AgriPerceiver is the only model that produces usable predictions across all six classes, lifting the three hardest classes from $\leq$ 0.025 to 0.525–0.560.

Table 3 reports the full precision/recall breakdown for AgriPerceiver. The row-normalised confusion matrix (Fig. 3) exposes the dominant error modes: *viral*→*deficiency* (29%) and *unknown*→*deficiency* (11%) both stem from diffuse chlorotic yellowing that is visually shared across viral, abiotic, and nutrient aetiologies, while *bacterial*→*fungal* (21%) and *pest*→*fungal* (23%) reflect the morphological overlap between early lesions and fungal necrosis at 384 × 384 tile resolution. These confusions, rather than structural failure, dominate the residual error and point to clear targets for data augmentation. In absolute terms the largest single off-diagonal cell is *unknown*→*deficiency* (169 of 1,535), while the strongest confusion by row fraction is *viral*→*deficiency* (101/345 = 29.3%),

Table 3. Per-class precision, recall, and F1 for AgriPerceiver on the 5,062-sample test set, with support counts. Macro- and support-weighted averages are shown; micro-averaged accuracy is 0.715.

Class	Support	Precision	Recall	F1
Fungal	1,521	0.790	0.838	**0.813**
Unknown	1,535	0.908	0.759	**0.827**
Bacterial	692	0.607	0.520	0.560
Deficiency	618	0.528	0.707	0.605
Pest	351	0.509	0.578	0.541
Viral	345	0.523	0.528	0.525
Macro avg	5,062	0.644	0.655	**0.645**
Weighted avg	5,062	0.731	0.715	0.719

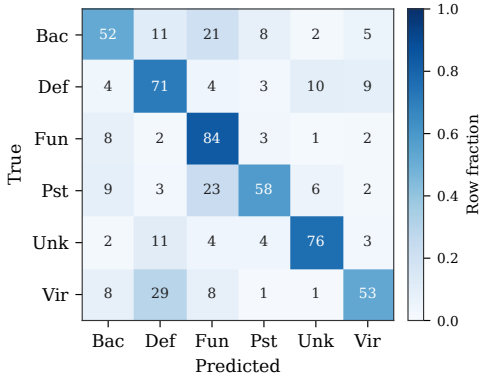


Fig. 3. Row-normalised confusion matrix for AgriPerceiver pathology-type prediction (cells show row percentages; n=5,062). Class order: Bac(terial), Def(iciency), Fun(gal), P(e)st, Unk(nown), Vir(al). The strong diagonal is punctuated by chlorosis- and necrosis-driven confusions (*e.g.* viral→deficiency, pest→fungal).

followed by *pest→fungal* (80/351 = 22.8%) and *bacterial→fungal* (145/692 = 21.0%).

6.4 Calibration

Because a severity/diagnosis report is only actionable if its confidence is trustworthy, we examine calibration explicitly. AgriPerceiver is moderately overconfident, with an ECE of 0.139. The gap is largest in the mid-confidence $[0.7, 0.8]$ bin (mean confidence 0.797 vs. accuracy 0.571) and tightens in the top decile (mean confidence 0.982 vs. accuracy 0.912). This inherited overconfidence is structurally expected, since the training targets carry Gemma-3's own confidence, and is directly addressable by post-hoc temperature scaling, which we leave to future work. Notably, AgriPerceiver is far better calibrated than InternVL2 and Qwen2-

Table 4. Runtime ablations on the full 5,062-sample test set (no retraining). ↑: higher is better; ↓: lower is better.

Variant	F1↑	FM↑	MAE↓	Comp.↑	Δ
Full model	0.645	0.486	0.067	0.810	—
− AnyRes	0.638	0.480	0.069	0.806	−0.004
− Tile embed.	0.647	0.483	0.067	0.810	≈ 0

VL (ECE ≈ 0.38), whose confidence collapses toward 0.9 regardless of correctness; LLaVA-NeXT's near-zero ECE is a degenerate artefact of its constant-"unknown" strategy, not genuine calibration. Only five of the ten bins are non-empty and just 0.3% of samples fall below 0.6 confidence; the $[0.7, 0.8]$ bin alone (n=1,354, gap 0.226) contributes 0.061 of the total ECE.

6.5 Ablations

Two runtime ablations isolate the spatial processing components by modifying inference without retraining (Table 4). Removing AnyRes (replacing all five tiles with five copies of the global view) reduces composite from $0.810 \rightarrow 0.806$ (Δ=−0.004), with type macro-F1 falling from $0.645 \rightarrow 0.638$ and severity MAE rising from $0.067 \rightarrow 0.069$, confirming that quadrant tiling provides spatial diversity that aids fine-grained lesion localisation. Zeroing tile-position embeddings $\mathbf{E}$ yields no composite change ($\Delta \approx 0$), though diagnosis fuzzy-match falls 0.003, confirming that spatial grounding aids free-text localisation without dominating the aggregate. Ablations requiring separate training (no-LoRA, latent counts 64/256) are deferred to future work.

6.6 Discussion

Three findings stand out. First, structural compliance is *not* a differentiator: all four systems emit valid JSON above 99.7%, so the meaningful gap lies entirely in the semantic and classification quality of the content, which prompt engineering alone does not close. Second, the value of domain specialisation concentrates on the hard, low-support classes: generalists handle the easy fungal/unknown majority but fail exactly where agronomic value is highest. Third, aggressive visual compression is compatible with strong fine-grained performance, provided spatial provenance is preserved through tile-position embeddings; the Perceiver bottleneck acts as a regulariser rather than an information sink. All of this is obtained while training only 9.2% of the parameters (426M of 4.65B; Appendix A) and while the 28.5× token compression cuts per-layer attention cost by 27.5× (Sect. 4.6), so high-resolution inference stays practical on a single GPU. Together these support parameter-efficient specialisation as a practical template for structured scientific reporting on commodity academic hardware.

7 Conclusion

We presented **AgriPerceiver**, achieving a composite score of **0.810** on structured agricultural pathology report generation. By leveraging a Perceiver Resampler to achieve a 28.5× visual token compression and a 27.5× reduction in per-layer attention cost, the model requires training on just 9.2% of its parameters, keeping the multi-billion-parameter SigLIP and Phi-3 backbones strictly frozen. AgriPerceiver establishes a computationally accessible, modular template for adapting foundation models to life-science domains requiring high-resolution visual fidelity and strict structural compliance.

Limitations and Future Work. Relying on Gemma-3-12B-IT for automated annotations carries the risk of inheriting teacher bias, and an ECE of 0.139 indicates moderate overconfidence (addressable via temperature scaling). The dominant residual errors are chlorosis- and necrosis-driven class confusions that call for targeted augmentation. Future work will prioritise acquiring agronomist-validated labels, developing longitudinal multi-image tracking to monitor disease progression, and exploring cross-crop transfer learning.

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A Implementation Details

Table 5 gives the exact parameter count of every trainable component, derived analytically from the architecture specification and verified against the Stage-1 checkpoint (`stage1_connector_weights.pt`: 745 MiB at bfloat16 = 390.6M parameters). Each PerceiverBlock contributes $4d^2 \approx 37.75$M parameters for cross-attention (query, key, value, and output projections at d=3072), the same again for self-attention, and 113.27M for the GLU-FFN (Linear(d, $8d$) gated projection plus Linear($4d$, d) down-projection). The LoRA adapters target five projection matrices in each of Phi-3's 32 transformer layers; at rank r=32 they add ∼35M updatable parameters, that is 0.92% of the language model's weights.

Table 5. Component-level parameter budget (left) and training hyperparameters for both stages (right). The bridge comprises the three trainable sub-modules. Stage-2 gradient accumulation gives an effective batch size of $8 \times 2 = 16$, matching Stage 1; both stages use the same learning rate, selected by grid search over $\{10^{-5}, 5 \times 10^{-5}, 10^{-4}\}$.

Component	**Params**
SigLIP-SO400M (frozen)	∼400M
TileEmbeddings (5×1152)	5,760
VisionProjector (2-layer MLP)	12.98M
PerceiverResampler (2 blk.)	377.99M
Latents (128×3072)	393K
2× Cross-attention	75.50M
2× Self-attention	75.52M
2× GLU-FFN	226.55M
Bridge total (Stage 1)	**390.98M**
Phi-3-mini-128k (frozen)	∼3.82B
LoRA adapters (r=32)	∼35M
Total (Stage 2)	∼4.65B
Trainable Stage 1	**391M (8.4%)**
Trainable Stage 2	**426M (9.2%)**

Parameter	**Stage 1**	**Stage 2**
Optimiser	AdamW	AdamW
Learning rate	10^{-4}	10^{-4}
LR schedule	Cosine warm-up	
Warm-up steps	500	500
Weight decay	0.01	0.01
Grad. clip	1.0	1.0
Batch size/GPU	16	8
Grad. accum.	1	2 (eff. 16)
Epochs	2	3
Steps	14.7K	37.9K
LoRA rank r	—	32
LoRA α	—	64
LoRA dropout	—	0.1
Precision	bfloat16	bfloat16
Random seed	42	42
Hardware	NVIDIA H200	
Peak VRAM	∼20 GB	∼35 GB
Wall-clock time	∼6 h	∼18 h

B Diagnostic Report Schema and Example Output

The seven-field schema $\mathcal{S}$ of Eq. (1) is given below, followed by a representative prediction for a fungal-infected maize leaf (severity 0.62, correctly classified). Admissible values: `type` ∈ {fungal, bacterial, viral, pest, deficiency, unknown}; `severity` and `confidence` ∈ $[0, 1]$ (0 = none/uncertain, 1 = lethal/certain); all remaining fields are free-form natural language.

```
{ "diagnosis": string, "type": enum, "severity": float [0,1],
  "confidence": float [0,1], "symptoms": list[string],
  "reasoning": string, "recommended_actions": list[string] }
```

```
{
  "diagnosis": "Northern Corn Leaf Blight",
  "type": "fungal", "severity": 0.62, "confidence": 0.81,
  "symptoms": ["elongated tan lesions, wavy margins",
               "chlorotic halo extending 2-3mm",
               "parallel necrotic streaks on lamina"],
  "reasoning": "Elliptical pale-tan lesions parallel to the leaf
    axis; consistent with Exserohilum turcicum under high humidity.",
  "recommended_actions": ["Apply propiconazole fungicide within 48h",
                          "Improve canopy airflow, row thinning",
                          "Monitor neighbours every 3 days"]
}
```

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